

Mohana Hemanth Kundurthi

Data Scientist

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SUMMARY

- **Data Scientist** with **4+ years** of expertise in statistical analysis, machine learning, and deep learning using Python (**NumPy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch**) for predictive modeling, classification, clustering, and NLP.
- Proficient in data wrangling, cleaning, and manipulation with **SQL, Spark, and Pandas**; experienced in working with structured, semi-structured, and unstructured data from diverse sources including APIs, databases, and cloud storage.
- Hands-on experience with data visualization and dashboarding tools such as **Tableau, Power BI, Matplotlib, Seaborn, Plotly, and Looker** to deliver compelling insights and drive data-informed decision-making.
- Skilled in building and deploying end-to-end data pipelines using tools like **Apache Airflow, AWS (S3, Lambda, SageMaker), GCP (BigQuery, Dataflow), and Docker** for scalable data solutions and model deployment.
- Strong background in statistical modeling, **A/B testing, time series forecasting**, and **hypothesis testing** with tools like **SciPy, Statsmodels, and Excel** for robust business and scientific analyses.
- Experienced in **Agile development, version control (Git)**, and collaborative environments using Jupyter Notebooks, VS Code, GitHub, Jira, and Confluence, fostering cross-functional teamwork and continuous integration.

TECHNICAL SKILLS	
Programming Language:	Python, R, SQL, SAS
Packages:	Pandas, NumPy, Matplotlib, Seaborn, Plotly, ggplot2, Scikit-learn, TensorFlow, PyTorch, OpenCV
Data Visualization:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP, Macros)
Machine Learning & AI:	Supervised Learning, Unsupervised Learning, Reinforcement Learning, Neural Networks, Random Forest, Gradient Boosting, XGBoost, SVM (Support Vector Machines), K-Means Clustering, KNN (K-Nearest Neighbors), Deep Learning (CNN (Convolutional Neural Networks), RNN (Recurrent Neural Networks), LSTM, GANs (Generative Adversarial Networks))
Cloud Platforms:	Amazon Web Service (AWS), Azure, Google Cloud Platform (GCP)
Database:	MySQL, PostgreSQL, MongoDB, MS SQL Server
Data Engineering Tools:	Apache Spark, Apache Hadoop, Apache Kafka, ETL/ELT
Other Technical Skills:	SSIS, SSRS, SSAS, Docker, Kubernetes, GitLab CI/CD Jenkins, Informatica, Talend, MLFlow, Snowflake, Google Big Query, A/B Testing, Hypothesis Testing, Statistical Modeling, Regression Analysis, Time Series Analysis, KPIs, Accuracy, Precision, Recall, F1-Score, ROC Curve, AUC, Apache Hive, Data Quality and Governance, Natural Language Process, Big Data, Advance Analytics, Data Mining, Data Visualization, Data warehousing, Data transformation, Data Integrity, Data Cleansing, Data Quality Assurance, problem-solving, SDLC, Agile Development, Testing and Development
Version Control Tools:	Git, GitHub, SVN
Operating Systems:	Windows, Linux, Mac iOS

WORK EXPERIENCE

Data Scientist | One Main Financial, MD **Jan 2024 – Present**

- Led the development of an AI-powered personalized financial planning assistant using transformer-based models (GPT-3, T5) and fine-tuned LLMs to analyze customer spending behavior, categorize expenses, and provide tailored savings recommendations.
- Designed and implemented scalable ML pipelines using Python, PyTorch, MLflow, and Airflow, enabling efficient deployment and monitoring of NLP models in a cloud-native production environment.
- Built and maintained real-time semantic search systems using vector similarity and embeddings stores, improving customer query resolution rates by 38% across Capital One’s mobile app.
- Applied advanced NLP techniques including summarization, intent classification, and context-aware embeddings to enhance customer support automation and reduce resolution time.
- Partnered with product designers and engineers to create intuitive user experiences, ensuring smooth integration of AI insights into digital interfaces for mobile and web platforms.
- Conducted continuous model performance evaluations using custom metrics, incorporating reinforcement learning to refine recommendation strategies based on user interactions and feedback loops.

Data Scientist | **Tata Consultancy Services, India**

May 2021 – Dec 2022

- Developed a predictive model to identify patients at high risk of 30-day hospital readmission using Python, SQL, and Vertex AI, applying advanced statistical techniques and working with cross-functional teams to align clinical goals with model design.
- Engineered and processed over 20 million rows of structured healthcare claims and provider data in Snowflake, leveraging feature selection, data normalization, and imputation strategies to enhance model performance and reliability.
- Collaborated with clinicians and product stakeholders to incorporate predictive insights into care coordination workflows, which contributed to a 12% increase in preventive care adherence and improved patient engagement rates.
- Designed, trained, and validated machine learning models including Logistic Regression, XGBoost, and Random Forest, selecting the most interpretable solution with 85% accuracy and achieving a 25% reduction in false positives for care gap predictions.
- Implemented an automated ML pipeline using Vertex AI Pipelines and Cloud Composer, reducing manual intervention and cutting deployment time by 80%, enabling consistent model updates and scalable execution.
- Translated model insights and performance metrics into executive-ready visualizations using Tableau and presentation decks, influencing leadership decisions and extending the project's impact to over 100,000 members across the care delivery ecosystem.

Data Analyst | **Wipro, India**

May 2019 – April 2021

- Processed and analyzed 1M+ rows of structured and unstructured data using SQL, Python (Pandas, NumPy), and Tableau for business intelligence reporting.
- Developed automated dashboards in Power BI and Tableau, reducing manual reporting efforts by 70%.
- Built predictive models for sales forecasting and customer churn analysis, improving accuracy by 25%.
- Conducted A/B testing and statistical analysis to identify key revenue growth opportunities, leading to an 8% increase in revenue.

EDUCATION

Master of Professional Studies in Data Science

University of Maryland Baltimore County, MD, USA

Bachelor of Technology in Mechanical Engineering

RVR & JC College of Engineering

PROJECTS

Visual Question Answering (VQA) for Accessibility

- Designed an AI-powered data annotation pipeline to preprocess VQA 2.0 and VizWiz datasets, improving model accuracy by 54.3%.
- Performed data cleaning and feature engineering to optimize model inference time by 10%.
- Integrated speech-to-text insights, reducing manual data dependency by 60%.

Object Detection for Boston Dynamics Spot Robot

- Developed a real-time data pipeline for object detection and tracking, improving spatial estimation efficiency by 40%.
- Implemented a Flask API to stream live camera footage across multiple devices, enabling remote monitoring and analysis.
- Conducted exploratory data analysis (EDA) on multi-sensor data to fine-tune YOLOv8 detection models.

Line and Area Counting using StrongSORT

- Built a data-driven interface for real-time person counting and spatial analysis, enhancing business intelligence for security analytics.
- Optimized tracking algorithms, improving accuracy by 25% in real-world conditions.

SLAM-Based Person Localization in Known Layout

- Mapped real-time positions with 2 cm error margins using LiDAR and camera data, enabling precise location analytics.
- Visualized movement heatmaps using Matplotlib and Seaborn, improving operational insights for facility management.

Perfect Pallet for Inventory Management Project

- Cleaned, prepared and annotated datasets for YOLOv5 training, achieving 79.5% accuracy and implementing robust performance tracking to ensure model effectiveness.
- Performed exploratory data analysis (EDA) on inventory datasets, identifying patterns that improved stock placement and reduced mismanagement by 20%.